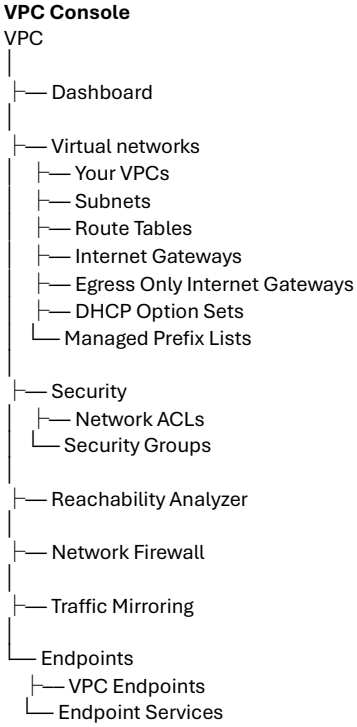
Amazon VPC — Console Dashboard Wise Important Interview Questions & Answers

VPC Console



---

1. What is Amazon VPC?

Answer:

Amazon VPC stands for Virtual Private Cloud. It allows us to create a logically isolated network inside AWS.

Inside a VPC, we can create subnets, route tables, security groups, internet gateways, and other networking components.

It gives us control over how our AWS resources communicate with each other and with the internet.

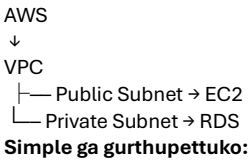
Example:

Suppose our application runs on EC2 and our database runs on RDS.

We can place both inside a VPC and control their communication using subnets, routes, and security groups.

This gives us a private and controlled AWS network.

Diagram:



Simple ga gurthupettuko:

VPC = AWS lo mana own private network

---

2. What do we see in the VPC Dashboard?

Answer:

The VPC Dashboard gives us an overview of the networking resources configured in our AWS account.

From here, we can quickly see VPCs, subnets, route tables, internet gateways, and other important resources.

It is mainly a starting point for managing the VPC environment.

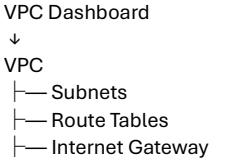
Example:

Suppose we create a VPC for our application.

From the VPC Dashboard, we can check the VPC and related networking components.

From there, we can open individual resources and configure them.

Diagram:



Simple ga gurthupettuko:

Dashboard = VPC environment overview

---

3. What is a VPC CIDR block?
Answer:
A CIDR block defines the IP address range available inside a VPC.
When we create a VPC, we assign a CIDR such as 10.0.0.0/16.
Subnets are then created using smaller CIDR ranges from the VPC CIDR.
Example:
Suppose our VPC uses 10.0.0.0/16.
We can create 10.0.1.0/24 for a public subnet and 10.0.2.0/24 for a private subnet.
These subnet ranges come from the VPC's IP range.
Diagram:
VPC
10.0.0.0/16
│
├── Public Subnet
│ └── 10.0.1.0/24
│
└── Private Subnet
└── 10.0.2.0/24
Simple ga gurthupettuko:
CIDR = VPC/Subnet IP range

---

4. What is a Subnet?
Answer:
A subnet is a smaller network range created inside a VPC.
A subnet belongs to one Availability Zone and is used to place AWS resources such as EC2 instances.
We commonly divide a VPC into public and private subnets based on the application's requirements.
Example:
Suppose our VPC has two Availability Zones.
We can create public subnets for load balancers and private subnets for application servers or databases.
This helps organize resources and control network access.
Diagram:
VPC
│
├── AZ-1
│ ├── Public Subnet
│ └── Private Subnet
│
└── AZ-2
├── Public Subnet
└── Private Subnet
Simple ga gurthupettuko:
Subnet = VPC lo smaller network

---

5. What is a Public Subnet?
Answer:
A public subnet is a subnet whose route table has a route to an Internet Gateway.
Resources in the subnet can communicate with the internet when they also have the required public addressing and security configuration.
Public subnets are commonly used for internet-facing resources such as load balancers.
Example:
Suppose we have an Application Load Balancer that must receive traffic from users on the internet.
We place the ALB in public subnets with a route to the Internet Gateway.
Users can then reach the load balancer through the internet.
Diagram:
Internet
↓
Internet Gateway
↓
Public Subnet
↓
ALB
Simple ga gurthupettuko:
Public Subnet = Internet route unna subnet

---

6. What is a Private Subnet?
Answer:
A private subnet does not have a direct route to an Internet Gateway.
Resources inside it can still access the internet through a NAT Gateway when outbound internet access is required.
Private subnets are commonly used for application servers and databases.

Example:
Suppose our RDS database should not be directly reachable from the internet.
We place it in private subnets and allow only the required application traffic.
This provides better network isolation.
Diagram:
Internet
X
|
Private Subnet
↓
RDS
Simple ga gurthupettuko:
Private Subnet = Direct internet route ledu

---

7. What is an Internet Gateway?
Answer:
An Internet Gateway, or IGW, allows communication between resources in a VPC and the public internet.
It must be attached to the VPC and referenced by a route table for internet-bound traffic.
It is commonly used with public subnets.
Example:
Suppose an ALB is running in a public subnet.
The subnet route table sends internet traffic to the Internet Gateway.
This allows internet users to communicate with the load balancer.
Diagram:
Internet
↓
Internet Gateway
↓
VPC
↓
Public Subnet
↓
ALB
Simple ga gurthupettuko:
IGW = VPC ↔ Internet connection

---

8. What is a Route Table?
Answer:
A route table contains rules that decide where network traffic should go.
Each route has a destination CIDR and a target such as an Internet Gateway, NAT Gateway, or another network component.
Subnets are associated with route tables to determine their traffic paths.
Example:
Suppose a public subnet needs internet access.
Its route table can contain 0.0.0.0/0 → Internet Gateway.
This tells traffic destined for the internet to use the Internet Gateway.
Diagram:
Public Subnet
↓
Route Table
↓
0.0.0.0/0
↓
Internet Gateway
↓
Internet
Simple ga gurthupettuko:
Route Table = Traffic ekkadiki vellalo decide chestundi

---

9. What is the difference between Main and Custom Route Table?
Answer:
Every VPC has a main route table that is used by subnets that are not explicitly associated with another route table.
A custom route table can be created when we need different routing behavior for specific subnets.
In production, we commonly create separate route tables for public and private subnets.
Example:
Suppose we have public and private subnets.
The public route table can point internet traffic to an Internet Gateway, while the private route table can point outbound traffic to a NAT Gateway.
This gives different routing behavior to the two subnet types.
Diagram:
VPC
├— Public Subnet → Public Route Table → IGW

└── Private Subnet → Private Route Table → NAT
Simple ga gurthupettuko:
Route Table = Subnet traffic rules

---

10. What is a NAT Gateway?
Answer:
A NAT Gateway allows resources in private subnets to initiate outbound connections to the internet.
It does not normally allow unsolicited inbound internet connections to those private resources.
A NAT Gateway is placed in a public subnet and uses an Internet Gateway for internet connectivity.
Example:
Suppose an EC2 server is in a private subnet and needs to download software updates.
Its traffic can go through a NAT Gateway in a public subnet.
The EC2 server can access the internet without becoming directly reachable from the internet.
Diagram:
Private EC2
↓
Private Route Table
↓
NAT Gateway
↓
Internet Gateway
↓
Internet
Simple ga gurthupettuko:
NAT Gateway = Private → Internet outbound

---

11. Why is NAT Gateway placed in a Public Subnet?
Answer:
A NAT Gateway needs connectivity to the internet through an Internet Gateway.
Therefore, it is placed in a public subnet whose route table has a route to the Internet Gateway.
Private subnet resources then send their internet-bound traffic to the NAT Gateway.
Example:
Suppose EC2 is in a private subnet.
The private route table sends 0.0.0.0/0 to the NAT Gateway.
The NAT Gateway then uses the Internet Gateway to reach the internet.
Diagram:
Private Subnet
↓
NAT Gateway
↓
Public Subnet
↓
IGW
↓
Internet
Simple ga gurthupettuko:
NAT = Public subnet lo untundi

---

12. What is a Security Group?
Answer:
A Security Group is a virtual firewall attached to resources such as EC2 instances.
It controls inbound and outbound traffic using rules based on protocols, ports, and source or destination.
Security groups are stateful, so return traffic is automatically allowed for permitted connections.
Example:
Suppose an EC2 web server should accept HTTP traffic on port 80 and SSH on port 22 only from a specific source.
We can create security group rules for those ports.
All other unwanted traffic can be blocked by not allowing it.
Diagram:
Internet
↓
Security Group
├── 80 → Allow
└── 22 → Allow from trusted source
↓
EC2
Simple ga gurthupettuko:
Security Group = Resource-level firewall

---

13. What is a Network ACL?
Answer:
A Network ACL, or NACL, is a firewall at the subnet level.

It controls inbound and outbound traffic for resources in the associated subnet.
NACLs are stateless, so return traffic must be explicitly allowed.
Example:
Suppose we want to control traffic entering and leaving an entire subnet.
We can use a NACL to allow or deny traffic based on IP addresses, protocols, and ports.
This gives us an additional network security layer.
Diagram:
Internet
↓
NACL
↓
Subnet
↓
EC2
Simple ga gurthupettuko:
NACL = Subnet-level firewall

---

14. Security Group vs NACL — What is the difference?
Answer:
A Security Group works at the resource level, while a NACL works at the subnet level.
Security Groups are stateful, whereas NACLs are stateless.
Security Groups mainly use allow rules, while NACLs support both allow and deny rules.
Example:
Suppose an EC2 instance is inside a subnet.
The NACL can control traffic for the whole subnet, while the EC2 Security Group can provide more specific access control for that instance.
We can use both together for layered security.
Diagram:
Internet
↓
NACL
↓
Subnet
↓
Security Group
↓
EC2
Simple ga gurthupettuko:
NACL = Subnet
SG = Resource

---

15. What is a VPC Peering Connection?
Answer:
VPC Peering allows two VPCs to communicate privately using private IP addresses.
The VPCs can be in the same or different AWS Regions, depending on the supported configuration.
The required routes must be added to the route tables on both sides.
Example:
Suppose one VPC contains the application and another VPC contains shared services.
We can create VPC Peering between them and configure routes.
The application can then communicate privately with the shared services.
Diagram:
VPC-A
EC2
↓
VPC Peering
↓
VPC-B
Shared Service
Simple ga gurthupettuko:
VPC Peering = VPC ↔ VPC private connection

---

16. What is a VPC Endpoint?
Answer:
A VPC Endpoint allows resources inside a VPC to connect to supported AWS services without requiring traffic to go through the public internet.
This can improve security and reduce the need for internet-facing network paths.
There are different endpoint types, including Gateway and Interface endpoints.
Example:
Suppose an EC2 instance in a private subnet needs to access S3.
We can configure an S3 VPC endpoint so the traffic can reach S3 without requiring a NAT Gateway for that access path.
This is commonly used in private architectures.

Diagram:
Private EC2
↓
VPC Endpoint
↓
S3
Simple ga gurthupettuko:
VPC Endpoint = Private connection to AWS service

---

17. What is a Gateway Endpoint?
Answer:
A Gateway Endpoint provides private connectivity from a VPC to supported AWS services such as S3 and DynamoDB.
It is configured through route tables rather than network interfaces.
It is commonly used when private subnet resources need access to these services without NAT.
Example:
Suppose EC2 instances in a private subnet need to download objects from S3.
We can create an S3 Gateway Endpoint and associate it with the required route table.
The EC2 instances can then access S3 through the endpoint.
Diagram:
Private EC2
↓
Route Table
↓
Gateway Endpoint
↓
S3
Simple ga gurthupettuko:
Gateway Endpoint = Private S3/DynamoDB access

---

18. What is an Interface Endpoint?
Answer:
An Interface Endpoint uses Elastic Network Interfaces, or ENIs, inside your subnets to provide private access to supported AWS services or endpoint services.
It uses AWS PrivateLink technology and can be protected using security groups.
It is commonly used for services that are not supported through Gateway Endpoints.
Example:
Suppose an application in a private subnet needs to access a supported AWS service privately.
We create an Interface Endpoint in the subnet.
The application can communicate with the service through the endpoint's private network interface.
Diagram:
Private EC2
↓
Interface Endpoint
↓
ENI
↓
AWS Service
Simple ga gurthupettuko:
Interface Endpoint = ENI-based private connection

---

19. What is a VPC Endpoint Service?
Answer:
A VPC Endpoint Service allows an application or service provider to expose a service privately to other VPCs using AWS PrivateLink.
The service provider creates the endpoint service, and consumers create interface endpoints to connect to it.
This allows private service-to-service communication without public internet access.
Example:
Suppose Company A provides an internal payment service.
Company B wants to consume that service privately from its VPC.
Company A can expose the service through an Endpoint Service, and Company B can connect using an Interface Endpoint.
Diagram:
Consumer VPC
↓
Interface Endpoint
↓
PrivateLink
↓
Endpoint Service
↓
Provider VPC
Simple ga gurthupettuko:
Endpoint Service = Mana service ni private ga expose cheyadam

20. What is a DHCP Option Set?
Answer:
A DHCP Option Set provides network configuration information to instances in a VPC through DHCP.
It can specify settings such as domain name and DNS servers.
A VPC can be associated with a DHCP option set to control these network settings.
Example:
Suppose our company uses an internal DNS server.
We can configure the appropriate DHCP options so instances in the VPC receive the required DNS configuration.
This helps AWS resources resolve internal domain names.
Diagram:
VPC
↓
DHCP Option Set
↓
EC2
↓
DNS Configuration
Simple ga gurthupettuko:
DHCP Option Set = Instance network configuration

---

21. What is a Managed Prefix List?
Answer:
A Managed Prefix List is a collection of CIDR blocks that can be referenced in network rules.
Instead of repeatedly entering many IP ranges, we can maintain them as a prefix list and reference it from supported resources.
This makes network configuration easier to manage.
Example:
Suppose our company has many approved network CIDR ranges.
Instead of adding each CIDR separately to multiple rules, we can maintain them in a prefix list.
When the list changes, the associated configurations can use the updated list.
Diagram:
Managed Prefix List
├— 10.0.1.0/24
├— 10.0.2.0/24
└— 10.0.3.0/24
↓
Network Rules
Simple ga gurthupettuko:
Prefix List = Group of CIDR ranges

---

22. What is VPC Flow Logs?
Answer:
VPC Flow Logs capture information about network traffic going to and from network interfaces in a VPC.
They are useful for troubleshooting connectivity and investigating network traffic.
Flow logs can be sent to destinations such as CloudWatch Logs or Amazon S3.
Example:
Suppose an EC2 server cannot connect to another server.
We can check VPC Flow Logs to understand whether traffic was accepted or rejected.
This can help identify network or security configuration problems.
Diagram:
EC2
↓
Network Traffic
↓
VPC Flow Logs
↓
CloudWatch / S3
↓
Analysis
Simple ga gurthupettuko:
Flow Logs = Network traffic information

---

23. What is Reachability Analyzer?
Answer:
Reachability Analyzer is a VPC analysis tool that checks whether a network path exists between two resources.
It analyzes the configured network components such as route tables, security groups, and NACLs.
It helps us identify why a connection is reachable or not reachable.
Example:
Suppose an EC2 instance cannot connect to another resource.
We can use Reachability Analyzer to analyze the path between them.
It can help identify whether a route or security rule is preventing connectivity.

Diagram:
EC2-A
↓
Reachability Analyzer
↓
Routes + SG + NACL
↓
EC2-B
Simple ga gurthupettuko:
Reachability Analyzer = Network path check

---

24. What is an Elastic IP?
Answer:
An Elastic IP is a static public IPv4 address that can be allocated to your AWS account.
It can be associated with supported resources such as an EC2 instance or NAT Gateway.
Unlike a normal automatically assigned public IP, an Elastic IP can remain associated with your account until you release it.
Example:
Suppose an application needs a stable public IP address.
We can allocate an Elastic IP and associate it with the required resource.
The application can then use the stable public IP.
Diagram:
Elastic IP
↓
EC2 / NAT Gateway
↓
Internet
Simple ga gurthupettuko:
Elastic IP = Static public IPv4

---

25. What is an Egress-Only Internet Gateway?
Answer:
An Egress-Only Internet Gateway is used to provide outbound internet access for IPv6 traffic from resources in a VPC.
It allows outbound communication but prevents the internet from initiating IPv6 connections back to those resources.
It is mainly used with IPv6-enabled private subnets.
Example:
Suppose an application uses IPv6 and needs to download updates from the internet.
We can use an Egress-Only Internet Gateway for outbound IPv6 connectivity.
The application can initiate connections without becoming directly reachable from the internet.
Diagram:
Private IPv6 Resource
↓
Egress-Only IGW
↓
Internet
Simple ga gurthupettuko:
Egress-Only IGW = IPv6 outbound only

---

26. What is a Default VPC?
Answer:
A Default VPC is a VPC automatically created by AWS in supported Regions for an AWS account.
It comes with default networking components that make it easier to launch resources quickly.
For production architectures, organizations often create and manage their own VPC design instead of relying only on the default VPC.
Example:
Suppose we create an EC2 instance in a new AWS account.
AWS may provide a Default VPC where the instance can be launched without manually creating all networking components first.
This is convenient for basic testing and learning.
Diagram:
AWS Region
↓
Default VPC
├— Subnets
├— Route Table
├— IGW
└— Security Group
Simple ga gurthupettuko:
Default VPC = AWS-provided ready network

---

27. What is a VPC Peering vs VPC Endpoint?
Answer:
VPC Peering connects one VPC to another VPC so resources can communicate privately.

A VPC Endpoint provides private access from a VPC to supported AWS services or endpoint services.
So Peering is mainly VPC-to-VPC communication, while Endpoint is service access.
Example:
Suppose VPC-A needs to communicate with VPC-B.
We can use VPC Peering for that connection.
If EC2 in VPC-A needs private access to S3, we can use a VPC Endpoint.
Diagram:
VPC Peering:
VPC-A ↔ VPC-B

VPC Endpoint:
VPC-A → Endpoint → AWS Service
Simple ga gurthupettuko:
Peering = VPC ↔ VPC
Endpoint = VPC → AWS Service

---

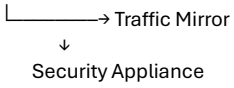
28. What is VPC DNS Resolution?
Answer:
VPC DNS resolution allows resources inside a VPC to resolve domain names to IP addresses.
VPC settings such as DNS resolution and DNS hostnames affect how DNS works for resources in the VPC.
This is important for services such as EC2, RDS, and internal application communication.
Example:
Suppose an EC2 instance needs to connect to an RDS database using its DNS endpoint.
The VPC's DNS functionality helps the EC2 instance resolve that hostname.
The application can then connect to the resolved database address.
Diagram:
EC2
↓
VPC DNS
↓
RDS DNS Name
↓
RDS IP
Simple ga gurthupettuko:
VPC DNS = Internal name → IP resolution

---

29. What is VPC Network Firewall?
Answer:
AWS Network Firewall is a managed network security service that provides deeper traffic inspection and filtering for VPC traffic.
It can be used for centralized network protection and more advanced firewall requirements.
It is different from Security Groups and NACLs because it provides additional inspection and firewall capabilities.
Example:
Suppose an organization needs to inspect and control traffic between different network areas.
It can use AWS Network Firewall to apply centralized firewall rules.
This adds another security layer to the VPC architecture.
Diagram:
Traffic
↓
Network Firewall
↓
VPC
↓
Application
Simple ga gurthupettuko:
Network Firewall = Advanced VPC traffic protection

---

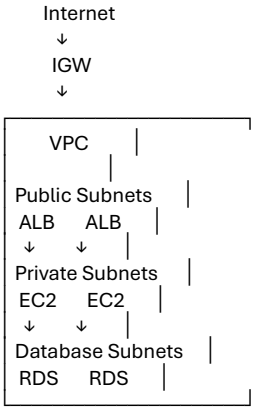
30. What is Traffic Mirroring?
Answer:
VPC Traffic Mirroring allows us to copy network traffic from supported network interfaces for security or troubleshooting purposes.
The mirrored traffic can be sent to monitoring or security appliances.
It is useful when we need deeper analysis of network traffic.
Example:
Suppose a security team wants to inspect traffic going to an EC2 application.
Traffic Mirroring can copy the traffic to a monitoring appliance.
The original application traffic can continue normally while the copied traffic is analyzed.
Diagram:
EC2
↓
Network Traffic
├————→ Application
│



Simple ga gurthupettuko:
Traffic Mirroring = Copy network traffic for inspection

---

31. What is the basic VPC architecture used in production?
Answer:
A common production VPC design uses multiple Availability Zones for high availability.
Public subnets usually contain internet-facing resources such as load balancers, while application and database resources are placed in private subnets.
Route tables, Security Groups, NACLs, NAT Gateways, and other components control communication.
Example:
Suppose we have a web application with an ALB, EC2 application servers, and RDS.
The ALB can be placed in public subnets, EC2 in private subnets, and RDS in separate private database subnets.
This provides a more secure and highly available architecture.
Diagram:



Simple ga gurthupettuko:
Production VPC = Public ALB + Private App + Private DB + Multiple AZs

---